SIS SAMPLE DRAFT — FULL TIER

Miami-Dade County Sample — Shopping Center / Retail

Traffic Impact Study

Prepared August 5, 2026 · 25.8195, -80.3690

SIS sample draft — full tier

Traffic Impact Study

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1.0 EXECUTIVE SUMMARY

This Traffic Impact Study has been prepared to evaluate the potential traffic impact, identify short-term roadway and circulation needs, determine potential mitigation strategies, and identify critical traffic issues that should be addressed during the planning process of the proposed Miami-Dade County Sample — Shopping Center / Retail, located at Custom site (25.8195, -80.3690), Miami-Dade County, within Miami-Dade County, Florida. The host controlling jurisdiction is Miami-Dade County (FDOT District 6 (Miami)); the applicable review framework is Concurrency retained + Chapter 33E multimodal mobility impact fee. Analysis follows the FDOT Multimodal Transportation Site Impact Handbook (MTSIH, March 25, 2024) and the FDOT Quality/Level of Service Handbook v6.0 (August 2025); capacity analysis follows the Highway Capacity Manual 6th Edition consistent with FDOT Traffic Analysis Handbook §4.1. The study covers 11 intersections within a 0.50-mile study area for land use 820 (Shopping Center / Retail) at a development size of 85 1,000 sqft GLA.

As part of this Traffic Impact Study, the following assignments were prepared consistent with the FDOT-approved methodology:

- Existing geometric conditions and assessment of the study impact area.
- Traffic data collection — daily volume counts and turning-movement counts at the critical study intersections and site driveways.
- Evaluation of existing traffic operations — Level of Service (LOS) and concurrency analysis.
- Traffic growth analysis, including committed-development assessment.
- Simulation of existing (current-year) and future conditions (opening year 2027) across three scenarios.
- Queue, turn-lane, and preliminary signal-warrant evaluation of the impacted network.

Findings:

• 0 intersections project to drop one or more LOS grades under Build conditions.

• 4 intersections operate at LOS E or F under Build conditions; mitigation per MTSIH 2024 §5 should be evaluated.

11	0	4	0.9s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

2.0 ANALYSIS METHODOLOGY

Per MTSIH 2024 §4.3, methodology and scope are established through a pre-application methodology meeting with the controlling FDOT District, county, and applicable MPO/TPO. For Miami-Dade County the pre-application methodology meeting is a hard prerequisite under the controlling local procedure (not an option); the meeting must occur and the methodology letter must be on file with the reviewing agency before this analysis can be accepted for review. The methodology letter or meeting minutes must be included as Appendix C of the formal submittal (Miami-Dade observed convention; canonical Florida default is Appendix A).

2.1 Controlling Guidance

Primary references: FDOT Multimodal Transportation Site Impact Handbook (MTSIH), March 25, 2024; FDOT Multimodal Transportation Site Impact Applications Guide, June 5, 2024; FDOT Quality/Level of Service Handbook v6.0, August 2025; FDOT Policy 000-525-006 (Level of Service Targets for the State Highway System); FDOT Procedure 525-030-120 (project traffic forecasting); FDOT Traffic Analysis Handbook (TAH), October 2025.

2.2 Analysis Software

Per FDOT TAH §4.1, approved analysis tools are HCS, Synchro / SimTraffic, SIDRA INTERSECTION (roundabouts), CORSIM, and Vissim. This screening analysis applies the HCM 6th Edition signalized-intersection model consistent with HCS output formatting. Vistro is not included in the FDOT TAH tool inventory; formal submittal output should be prepared in HCS or Synchro.

2.3 Traffic Data Collection

Per MTSIH 2024, roadway-segment counts should be 72 consecutive hours (Monday afternoon through Friday morning) in urbanized, transitioning, and urban area classes, and 7 days in rural areas, in 15-minute increments on typical weekdays excluding holiday weeks. The default analysis peak per MTSIH 2024 §2.3.1 is the Weekday PM Peak Hour of Adjacent Street Traffic (one hour between 4–6 PM); MTSIH 2024 imposes no blanket Saturday-peak requirement for retail or restaurant land uses — midday, Saturday, or other special peaks are added only where site characteristics warrant (the Applications Guide fast-food case study analyzes AM + PM + midday). For turning-movement counts, MTSIH 2024 Appendix A (p. A-3) requires AM and PM TMCs covering trucks, pedestrians, and bicycles but does not prescribe duration or bin size; both are agreed at the pre-application methodology meeting. The Applications Guide Case Study 2 (§3.4.3) uses 8-hour TMCs (3 hr AM + 2 hr midday + 3 hr PM) as a worked example, while 2-hr AM and 2-hr PM in 15-minute bins is common Florida practice.

2.4 Analysis Scenarios and Time Horizons

Per MTSIH 2024 §4.3, minimum analysis years are: Existing, Future Background (No-Build), Future Build, and Future Build with Mitigation. This study is organized around three scenarios consistent with FDOT District practice: Scenario 1 — Existing Conditions (current-year); Scenario 2 — Future Conditions No-Build (opening year 2027); and Scenario 3 — Future Conditions Build (opening year 2027). Opening year is canonical; there is no fixed +5 horizon for concurrency or connection-permit work. Each year is explicitly labeled. For a Comprehensive Plan Amendment (CPA) review, the analysis must additionally include short-term (5-year) and long-term (10-year minimum) horizons. Miami-Dade County convention: Per CDMP review; large CDMP applications typically use Existing + short-term build (e.g., 2020) + long-term build (e.g., 2040).

2.5 Trip Generation

Trip generation follows the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for land use 820 (Shopping Center / Retail) at the proposed development size of 85 1,000 sqft GLA. Net new external trips are calculated by applying pass-by and internal capture credits to gross trip generation per standard pass-by / internal-capture screening methodology. Per MTSIH 2024 §4.6.4, the fitted-curve equation is preferred when ≥20 data points are available, or when $R^2 \geq 0.75$ with the fitted curve falling within the data cluster and weighted standard deviation > 55% of the weighted average rate; otherwise the weighted average rate applies. Per MTSIH 2024 §4.6.6.6, pass-by trips at a site driveway cannot exceed 10% of the adjacent peak-hour two-way street traffic — this reasonableness check applies per roadway when the site fronts multiple streets and should be verified against the adjacent-street counts at submittal time.

Pass-by capture applied	30%
Internal capture applied	0% (MTSIH 2024 §4.6.9 sets no statewide numeric cap; rate negotiated at the methodology meeting per NCHRP 684 / standard screening methodology)
Background growth applied	1.5% per year over 1 year(s)
Miami-Dade County TIA threshold	Per Code Ch. 33-G concurrency procedures (Admin. Order 4-85); all concurrency-relevant trips reviewed
Weather condition	clear

Table: Site Trip Generation Summary

Period	Entering trips	Exiting trips
Daily	3,400	3,400
AM peak hour	166	106
PM peak hour	326	354

Table: Trip Generation by Period (raw → pass-by / internal capture → net external)

Period	Raw	Pass-by	Int. cap.	External	In	Out
AM Peak Hour	272	20	0	216	132	84
PM Peak Hour	680	204	0	408	196	212
Saturday Midday	748	56	0	593	296	297
Daily Total	6,800	510	0	5,390	2,695	2,695

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross − pass-by − internal-capture) × auto-mode share (85.7% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

2.6 Growth Rate

Per FDOT TAH §2.7, demand projections should use the adopted regional MPO/TPO travel-demand model (TDM); where TDM use is not warranted, historical AADT trend growth from Florida Traffic Online (FTO) is the FDOT-wide convention. Background traffic is grown at 1.5% per year over 1 year for this analysis; the rate should be confirmed against FDOT historical AADT and agreed upon during the methodology meeting.

2.7 Level of Service Standards

Per FDOT Policy 000-525-006, the peak-hour automobile-mode LOS standard on the State Highway System is LOS D in urbanized areas and LOS C in rural and transitioning areas. Constrained or backlogged facilities maintain their facility-specific designation. Roadway segment LOS reporting uses the FDOT Q/LOS Handbook v6.0 Generalized Service Volume Tables (GSVTs). Intersection LOS uses HCM 6th Edition Chapter 19 (signalized intersections), Exhibit 19-8 thresholds: A ≤10s, B ≤20s, C ≤35s, D ≤55s, E ≤80s, F >80s of average control delay per vehicle.

2.8 Context Classification

Per FDOT Q/LOS v6.0 (which replaced "complete streets" terminology with "context-based solutions") and FDM Chapter 200 §200.4 (Table 200.4.1), the study network's context classification (C1 Natural, C2 Rural, C2T Rural Town, C3R Suburban Residential, C3C Suburban Commercial, C4 Urban General, C5 Urban Center, C6 Urban Core) calibrates mode treatments and design standards; cross-section and lane widths follow FDM Table 210.2.1. The controlling context class should be confirmed against the FDOT Preliminary Context Classification mapping during the methodology meeting.

3.0 INTRODUCTION

The proposed Miami-Dade County Sample — Shopping Center / Retail is located at Custom site (25.8195, -80.3690), Miami-Dade County within Miami-Dade County, in the Miami-Dade County area. The project consists of 85 1,000 sqft GLA under ITE land use 820 (Shopping Center / Retail), and is anticipated to be completed by the opening year 2027.

Project name	Miami-Dade County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	25.8195°, -80.3690°
Region	Miami-Dade County
Host jurisdiction	Miami-Dade County
FDOT District	FDOT District 6 (Miami)
Review framework	Concurrency retained + Chapter 33E multimodal mobility impact fee
Controlling document(s)	CDMP + Administrative Order 4-85 + Code Ch. 33-G (concurrency) + Ch. 33E (mobility fee)
Regional MPO / TPO	Miami-Dade TPO
Opening year	2027

Surrounding land use, site plan figures, and detailed land-use description are dependent on the final site plan and are not produced by this screening tool. Final submittal should incorporate site plan figures (surrounding intersections and studied area) and a written project description per MTSIH 2024.

3.1 Study Area

The study area comprises the 11 intersections listed below, selected for their proximity to the site and their share of project-generated traffic. Existing lane geometry and traffic control at each location are shown in the Scenario 1 figures (see the turning-movement appendix).

#	Study intersection	Distance (mi)
1	NW 107 Av & NW 50 St	0.06
2	Northwest 107th Avenue & Northwest 52nd Street	0.16
3	NW 52 St @ 10400 Blk	0.22
4	NW 50 St @ NW 10800 Blk	0.28
5	NW 52 St @ NW 10400 Blk	0.30
6	NW 109 Av @ NW 4800 Blk	0.31
7	NW 52 St @ NW 104 Av	0.34
8	NW 50 St @ NW 10900 Blk	0.41
9	NW 107 Av & NW 58 St	0.46
10	NW 50 St @ NW 11000 Blk	0.46
11	NW 58 St @ NW 10400 Blk	0.49

STRUCTURE NOTE — MIAMI-DADE COUNTY CONVENTION

Miami-Dade CDMP-amendment submittals conventionally place the Engineer's Certification page as the FIRST section of the report (before the Executive Summary), and place the Methodology Letter at Appendix C rather than the canonical Appendix A. Confirm the Engineer of Record's seal and the Methodology Letter placement against the most recent submittal expectations before delivering.

Miami-Dade large-CDMP reports additionally conclude with three parallel-track end-chapters (Concurrency Analysis / CDMP Analysis / Zoning Analysis, each independently numbered 1.0–3.0). This screening tool does not auto-generate the three-track parallel content; §10.0 below identifies the concurrency inputs required.

4.0 SCENARIO 1 — EXISTING CONDITIONS

Scenario 1 establishes existing (current-year) conditions from the collected daily and turning-movement counts at the study intersections. Existing Level of Service and control delay are summarized below; existing lane geometry and traffic control are shown in the Scenario 1 turning-movement figures (appendix).

Study intersection	Distance (mi)	Existing LOS	Existing delay (s)
NW 107 Av & NW 50 St	0.06	F	120.0
Northwest 107th Avenue & Northwest 52nd Street	0.16	F	120.0
NW 52 St @ 10400 Blk	0.22	B	18.0
NW 50 St @ NW 10800 Blk	0.28	B	16.1
NW 52 St @ NW 10400 Blk	0.30	B	18.0

Study intersection	Distance (mi)	Existing LOS	Existing delay (s)
NW 109 Av @ NW 4800 Blk	0.31	B	16.1
NW 52 St @ NW 104 Av	0.34	B	18.0
NW 50 St @ NW 10900 Blk	0.41	B	16.1
NW 107 Av & NW 58 St	0.46	F	120.0
NW 50 St @ NW 11000 Blk	0.46	B	16.1
NW 58 St @ NW 10400 Blk	0.49	F	120.0

Existing AADT counts and functional classification are seeded from a live query of the FDOT Transportation Data and Analytics (TDA) ArcGIS REST services (gis.fdot.gov RCI_Layers FeatureServer 0/3/15 + Access_Management_TDA) within an 80-meter buffer of each intersection. Where the live extract returns a row, the in-table AADT, AADT year, and access-management class reflect the FDOT-published value at render time. Where no segment is matched (off-SHS local-road intersections, or radii beyond 80 m), values fall back to the engine estimate and must be confirmed against Florida Traffic Online (https://tdaappsprod.dot.state.fl.us/fto/). Existing turn-lane storage length must be field-supplied as `existingStorageFt` on each intersection record to enable the §9.0 storage-bay-adequacy comparison; the renderer surfaces a deficit calculation only when that field is present.

4.1 FDOT TDA Site Snapshot (live)

RCI roadway ID	87000509
Begin/end mileposts	0.000 → 1.000
AADT	30,000 vpd (2025)
K factor (peak-hour)	9.000
D factor (directional)	55.000
Truck %	8.7%
Functional class	Urban Minor Arterial
On State Highway System?	—
Access management class	Not classified — interim Rule 14-97.004(1) standards apply
FDOT District	D-6

Live extract from gis.fdot.gov/arcgis/rest/services/RCI_Layers/FeatureServer at render time. Nearest-segment match at 200 m (intersection point sits off the RCI polyline — likely in median or frontage road; verify route attribution). Cross-check against Florida Traffic Online (https://tdaappsprod.dot.state.fl.us/fto/) for the most recent annual update before submittal.

4.2 Roadway Segments Configuration

Physical and operational characteristics of each study segment from a live FDOT Transportation Data and Analytics (TDA) Roadway Characteristics Inventory (RCI) query at render time. Through-lane count and median type are RCI flat-file attributes not returned by this point query and must be confirmed from the RCI or a field inventory at submittal.

Study segment	RCI roadway	Functional class	SHS	Access cls	K	D
NW 107 Av & NW 50 St	87000509	Urban Minor Arterial	—	—	9.000	55.000
Northwest 107th Avenue & Northwest 52nd Street	87000509	Urban Minor Arterial	—	—	9.000	55.000
NW 52 St @ 10400 Blk	87000970	Urban Collector	—	—	9.000	55.000
NW 50 St @ NW 10800 Blk	87000973	Urban Local	—	—	9.000	55.000

Study segment	RCI roadway	Functional class	SHS	Access cls	K	D
NW 52 St @ NW 10400 Blk	87000970	Urban Collector	—	—	9.000	55.000
NW 109 Av @ NW 4800 Blk	87900018	Urban Local	—	—	—	—
NW 52 St @ NW 104 Av	87000970	Urban Collector	—	—	9.000	55.000
NW 50 St @ NW 10900 Blk	87000973	FUNCLASS 18	—	—	9.000	55.000
NW 107 Av & NW 58 St	87000953	Urban Minor Arterial	—	—	9.000	55.000
NW 50 St @ NW 11000 Blk	87000973	FUNCLASS 18	—	—	9.000	55.000
NW 58 St @ NW 10400 Blk	87000534	Urban Minor Arterial	—	—	9.000	55.000

Live extract from gis.fdot.gov RCI_Layers FeatureServer. Access-management class per Rule 14-97 F.A.C.; functional classification per RCI Feature 121. Confirm lane count, median type, and posted speed from the RCI flat file or a field inventory before submittal.

5.0 SCENARIO 2 — FUTURE CONDITIONS NO BUILD (2027)

Scenario 2 represents future background (No-Build) conditions at the opening year 2027: existing volumes grown to the opening year without the proposed project. Background traffic is grown at 1.5% per year over 1 year, and any committed developments in the study area should be added to the No-Build network. The methodology and derivation of the applied growth rate are described in §2.6; the resulting No-Build Level of Service at each study intersection is reported in §7.0 alongside the Existing and Build scenarios.

Where the reviewing agency requires a travel-demand-model forecast (e.g., SERPM for Southeast Florida, or the controlling MPO/TPO model), the No-Build volumes should be reconciled against the adopted model run; the model version, base year, and horizon year are identified in the methodology letter. This screening analysis applies historical-AADT trend growth per FDOT TAH §2.7 in lieu of a model run.

5.1 AADT Growth Projection

Current-year segment AADT (live FDOT RCI, §4.2) is projected to the No-Build horizons at the applied 1.50%/yr compound growth rate derived from the FDOT historical-AADT trend (§2.6). These are screening-level trend projections; the adopted MPO/TPO travel-demand model volumes govern at formal submittal.

Study segment	Current AADT (yr)	CAGR	No-Build AADT (2027)	Design AADT (2047)
NW 107 Av & NW 50 St	30,000 (2025)	1.50%	30,907	41,627
Northwest 107th Avenue & Northwest 52nd Street	30,000 (2025)	1.50%	30,907	41,627
NW 52 St @ 10400 Blk	7,900 (2025)	1.50%	8,139	10,962
NW 50 St @ NW 10800 Blk	5,000 (2025)	1.50%	5,151	6,938
NW 52 St @ NW 10400 Blk	7,900 (2025)	1.50%	8,139	10,962
NW 52 St @ NW 104 Av	7,900 (2025)	1.50%	8,139	10,962
NW 50 St @ NW 10900 Blk	5,000 (2025)	1.50%	5,151	6,938
NW 107 Av & NW 58 St	31,000 (2025)	1.50%	31,937	43,014

Study segment	Current AADT (yr)	CAGR	No-Build AADT (2027)	Design AADT (2047)
NW 50 St @ NW 11000 Blk	5,000 (2025)	1.50%	5,151	6,938
NW 58 St @ NW 10400 Blk	26,500 (2025)	1.50%	27,301	36,770

Projection = current AADT × (1 + CAGR)^(horizon − count year). Where the live RCI count year is unavailable, the applied growth-years span is used. For submittal, use the per-segment FDOT historical AADT series (Florida Traffic Online) and, where required, the adopted MPO/TPO model volumes.

5.2 SERPM Projected Model Volumes

Regional travel-demand-model volumes for the study corridors from the adopted Southeast Florida Regional Planning Model (SERPM 9.62, FDOT District 4), fetched at render time from the published FDOT D4 loaded-network layers. Values are the nearest model link's base-year (2020) and horizon-year (2050) daily and PM-peak volumes; confirm link-node correspondence to the specific study segments against the adopted model at submittal.

Study segment	Base 2020 daily	Base PM	Future 2050 daily	Future PM
Northwest 107th Avenue & Northwest 52nd Street	16,635	4,546	16,056	4,471
NW 52 St @ NW 10400 Blk	4,508	1,236	5,515	1,655
NW 109 Av @ NW 4800 Blk	3,036	605	3,826	1,291
NW 52 St @ NW 104 Av	4,508	1,236	5,515	1,655
NW 50 St @ NW 10900 Blk	3,036	605	3,826	1,291
NW 107 Av & NW 58 St	19,677	5,102	18,703	5,477
NW 50 St @ NW 11000 Blk	2,447	635	2,765	1,068
NW 58 St @ NW 10400 Blk	11,013	4,381	13,261	5,158

Source: FDOT D4 SERPM loaded networks (services1.arcgis.com … D4_Travel_Demand_Models, layers 0 and 7), nearest link within 200 m by highest daily volume. The SERPM horizon volumes are the authoritative No-Build growth basis where the reviewing agency requires a model forecast in lieu of historical-trend growth (§2.6 / §5.1).

6.0 SCENARIO 3 — FUTURE CONDITIONS BUILD (2027)

Scenario 3 represents future Build conditions at the opening year 2027: the Scenario 2 No-Build network plus the proposed development's net new external trips at the assigned distribution. Per FDOT TAH §2.7, trip distribution and assignment should use the adopted regional MPO/TPO travel-demand model (Miami-Dade TPO), with model version, base year, and horizon year identified in the methodology letter. This screening analysis distributes net new external trips to study-area zones with the Caltran gravity model (mass ÷ distance; see §6.1) and assigns them to signalized intersections within the study area; for formal submittal, distribution percentages and the TDM run identifier should be agreed upon during the methodology meeting.

6.1 Trip Distribution — Gravity Model

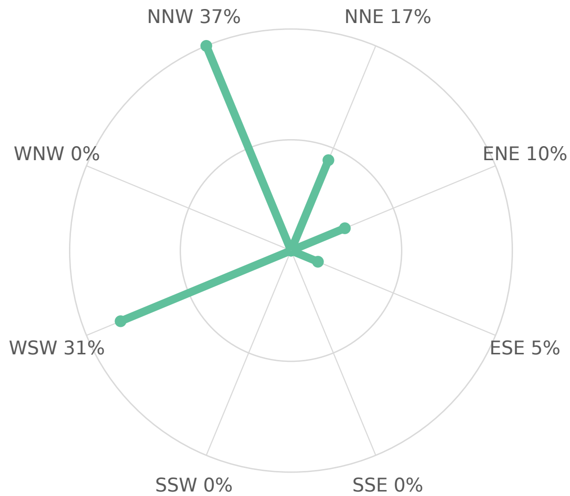
Project trips are distributed to the surrounding study-area zones with the gravity model used in the Caltran Engineering HCA Westside reference TIS, adopted here as the Florida distribution standard. Each zone attracts trips in proportion to its mass and inversely with its distance from the site — term = M / (d^1 · d_site) — normalized so the zone shares sum to 100%. Zone mass M is the intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy; d_site (the site zone's own distance normalizer) is 1. The resulting shares set the directional distribution below and drive the project-trip assignment in §6.2.

Directional sector	Share of project trips
Northeast (NNE + ENE)	26%
Southeast (ESE + SSE)	5%
Southwest (SSW + WSW)	31%
Northwest (WNW + NNW)	37%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
NW 107 Av & NW 50 St	WSW	0.06	2,700	2700.0	18.73%
Northwest 107th Avenue & Northwest 52nd Street	NNW	0.16	2,700	2700.0	18.73%
NW 107 Av & NW 58 St	NNW	0.46	2,700	2700.0	18.73%
NW 58 St @ NW 10400 Blk	NNE	0.49	2,385	2385.0	16.54%
NW 52 St @ 10400 Blk	ENE	0.22	711	711.0	4.93%
NW 52 St @ NW 10400 Blk	ENE	0.30	711	711.0	4.93%
NW 52 St @ NW 104 Av	ESE	0.34	711	711.0	4.93%
NW 50 St @ NW 10800 Blk	WSW	0.28	450	450.0	3.12%
NW 109 Av @ NW 4800 Blk	WSW	0.31	450	450.0	3.12%
NW 50 St @ NW 10900 Blk	WSW	0.41	450	450.0	3.12%
NW 50 St @ NW 11000 Blk	WSW	0.46	450	450.0	3.12%

Screening-grade gravity distribution over 11 study-area zones. For formal submittal the adopted regional MPO/TPO travel-demand-model distribution and the run identifier are confirmed at the methodology meeting per FDOT TAH §2.7; this gravity worksheet documents the screening basis.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length ∝ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

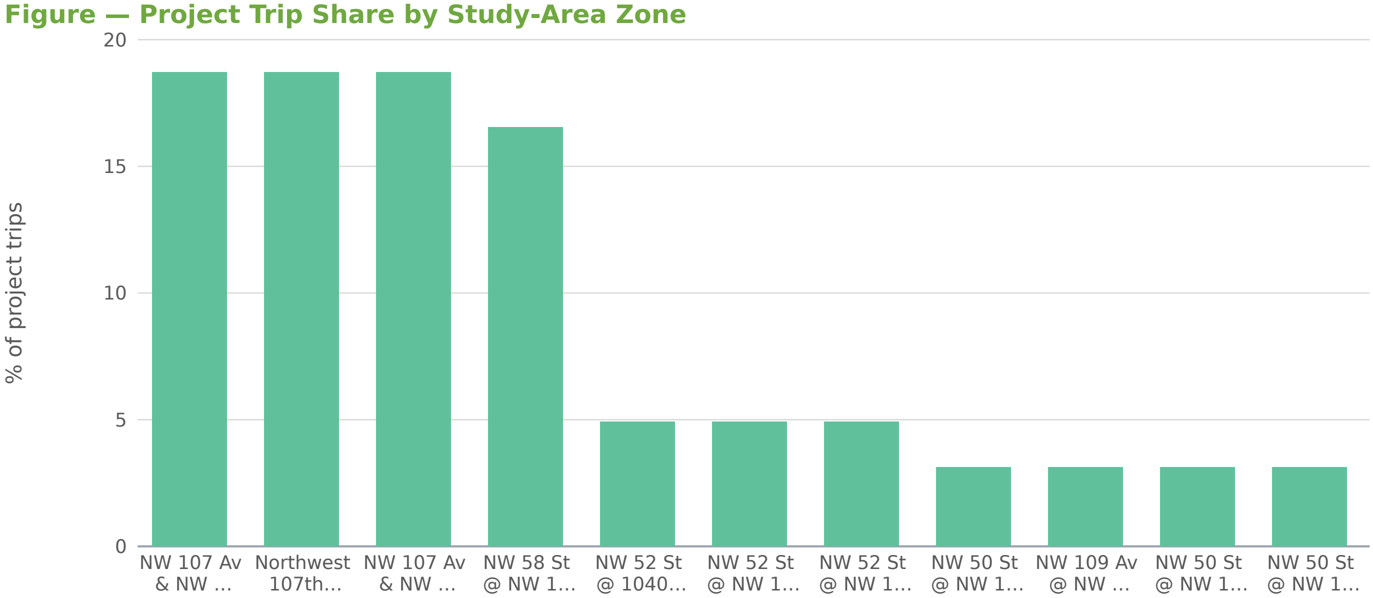
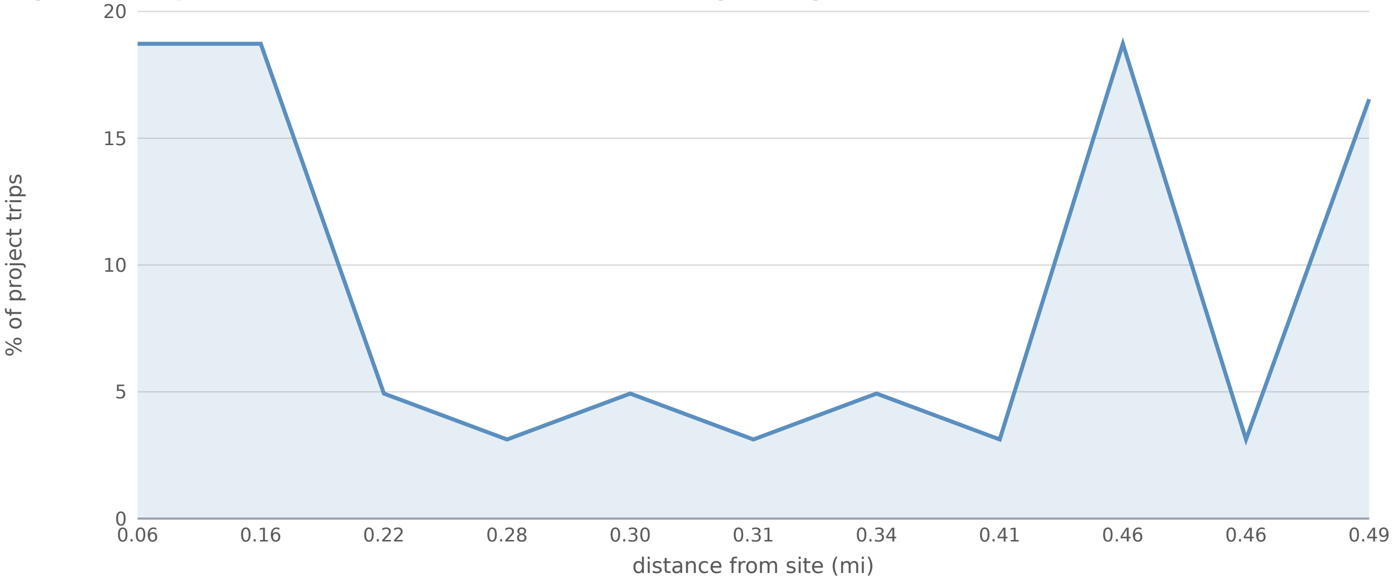


Figure — Trip Share vs. Distance from Site (Gravity Decay)



6.2 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
NW 107 Av & NW 50 St	WSW	0.06	202	382	31.8%
Northwest 107th Avenue & Northwest 52nd Street	NNW	0.16	148	280	23.3%
NW 52 St @ 10400 Blk	ENE	0.22	28	54	4.5%
NW 50 St @ NW 10800 Blk	WSW	0.28	67	127	10.6%
NW 52 St @ NW 10400 Blk	ENE	0.30	19	36	3.0%
NW 109 Av @ NW 4800 Blk	WSW	0.31	57	108	9.0%
NW 52 St @ NW 104 Av	ESE	0.34	8	15	1.2%
NW 50 St @ NW 10900 Blk	WSW	0.41	34	65	5.4%
NW 107 Av & NW 58 St	NNW	0.46	32	61	5.1%
NW 50 St @ NW 11000 Blk	WSW	0.46	27	51	4.2%
NW 58 St @ NW 10400 Blk	NNE	0.49	12	24	2.0%

AM- and PM-peak project trips assigned to each study intersection from the §6.1 gravity distribution: the directional shares orient the loading toward the high-share sectors, while distance-decay from the site sets the magnitude that passes through each intersection. Entering/exiting splits at each intersection follow the site's period directional split.

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

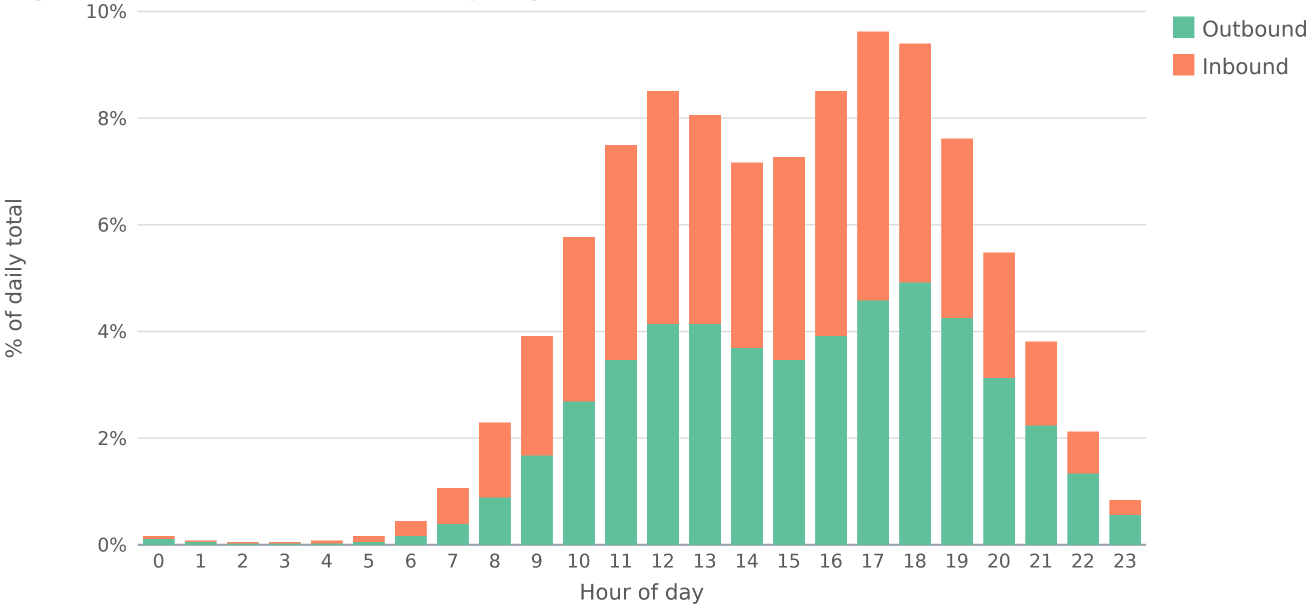
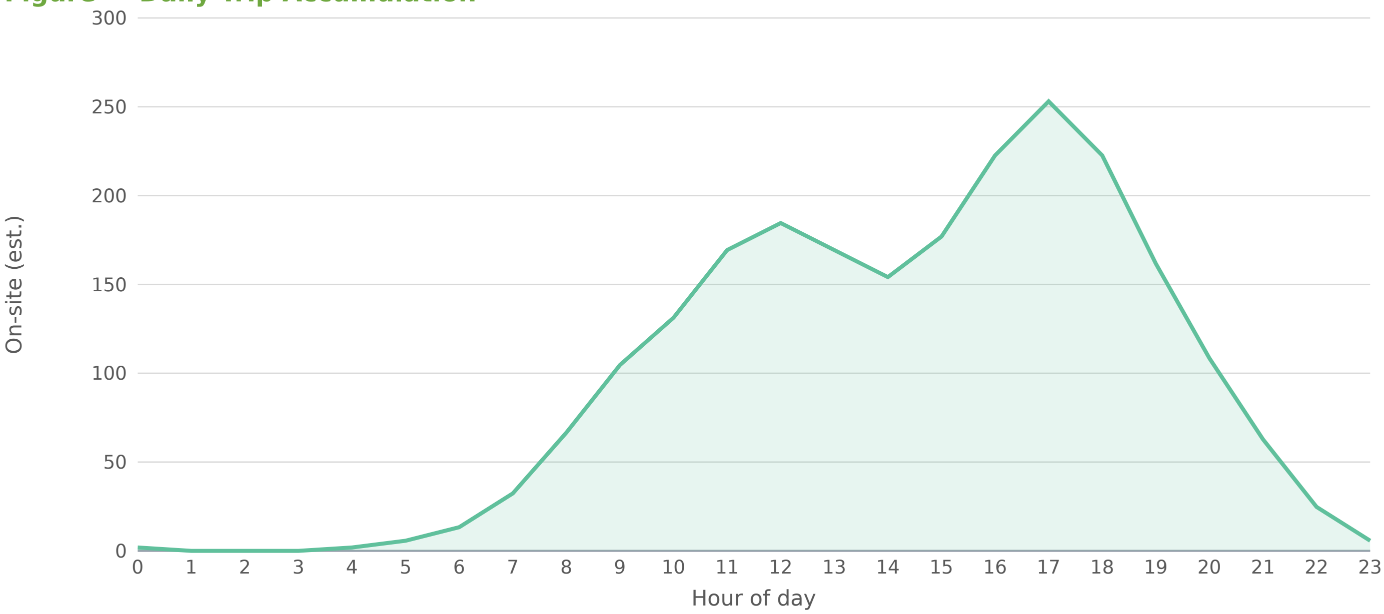


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

7.0 LEVEL OF SERVICE ANALYSIS

Intersection Level of Service is reported across the analysis scenarios per HCM 6th Edition Chapter 19 (Exhibit 19-8 control-delay thresholds). Four horizons are evaluated: (1) Existing — Scenario 1 current-year volumes; (2) No-Build — Scenario 2 at opening year 2027; (3) Build — Scenario 3 at opening year 2027; and (4) 20-Year Long-Range (2047) No-Build and Build. A ▲ flag marks any intersection projected to drop a LOS grade under Build conditions.

Intersection	Existing	Opening NB	Opening Bld	Design NB	Design Bld	Δ delay (s)
NW 107 Av & NW 50 St	F / 120	F / 120	F / 120	F / 120	F / 120	0.0
Northwest 107th Avenue & Northwest 52nd Street	F / 120	F / 120	F / 120	F / 120	F / 120	0.0
NW 52 St @ 10400 Blk	B / 18	B / 18	B / 19	C / 21	C / 21	0.5
NW 50 St @ NW 10800 Blk	B / 16	B / 16	B / 17	B / 17	B / 18	0.9
NW 52 St @ NW 10400 Blk	B / 18	B / 18	B / 18	C / 21	C / 21	0.3
NW 109 Av @ NW 4800 Blk	B / 16	B / 16	B / 17	B / 17	B / 18	0.8
NW 52 St @ NW 104 Av	B / 18	B / 18	B / 18	C / 21	C / 21	0.1
NW 50 St @ NW 10900 Blk	B / 16	B / 16	B / 17	B / 17	B / 18	0.5
NW 107 Av & NW 58 St	F / 120	F / 120	F / 120	F / 120	F / 120	0.0
NW 50 St @ NW 11000 Blk	B / 16	B / 16	B / 17	B / 17	B / 18	0.4
NW 58 St @ NW 10400 Blk	F / 120	F / 120	F / 120	F / 120	F / 120	0.0

Each scenario cell shows LOS grade / average control delay (s/veh) per HCM 6th Ed. Ex. 19-8 (A ≤10, B ≤20, C ≤35, D ≤55, E ≤80, F >80 s). Screening delays use a generic signal model (90 s cycle, g/C 0.45, 1,800 pc/h/ln, one critical lane per approach) and are superseded by a calibrated HCS/Synchro analysis of the actual lane geometry and signal timing at submittal.

7.1 Improvement Alternatives

The following intersection improvement alternatives are recommended to address projected Build-condition LOS impacts. Geometric mitigation should be designed to FDOT Design Manual (FDM 2026, Topic No. 625-000-002, dated January 1, 2026) standards — turn-lane warrants, deceleration / storage / taper lengths, intersection sight distance, and median-opening design per FDM Chapter 212; roundabouts per FDM Chapter 213. Proportionate-share, mobility-fee, or developer-contribution amounts for jurisdictions that retain concurrency (e.g., Miami-Dade Chapter 33-G) or operate mobility-fee programs (e.g., Hillsborough, Jacksonville/Duval Chapter 655, Miami-Dade Chapter 33E) should be calculated separately based on the controlling local-government ordinance.

NW 107 Av & NW 50 St [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Northwest 107th Avenue & Northwest 52nd Street [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

NW 107 Av & NW 58 St [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

NW 58 St @ NW 10400 Blk [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

7.2 Alternative Intersection LOS (with improvement)

The improvement alternatives in §7.1 are sized to mitigate the project's incremental impact and restore each affected intersection to approximately its No-Build (without-project) Level of Service. The table below states the with-improvement objective per location; the achieved Level of Service must be confirmed by a detailed HCS / Synchro analysis of the specific improvement at submittal.

Intersection	Build LOS	Δ delay (s)	Improvement class	Target LOS (improved)
NW 107 Av & NW 50 St	F	0.0	Major (geometry + signal)	F (verify — may remain constrained)
Northwest 107th Avenue & Northwest 52nd Street	F	0.0	Major (geometry + signal)	F (verify — may remain constrained)
NW 107 Av & NW 58 St	F	0.0	Major (geometry + signal)	F (verify — may remain constrained)
NW 58 St @ NW 10400 Blk	F	0.0	Major (geometry + signal)	F (verify — may remain constrained)

Target LOS is the No-Build (without-project) grade the improvement is sized to restore, not a re-run HCM result. Confirm the achieved delay / LOS with a detailed HCS / Synchro analysis of the specific geometry and signal timing at submittal.

8.0 QUEUE ANALYSIS (2027)

The 95th-percentile back-of-queue at each study intersection under Build conditions is summarized below (worst-approach basis). Queues are estimated from the HCM 6th Edition signalized-intersection model; a formal submittal should report per-lane-group queues from a Synchro / SimTraffic run and compare them to the available turn-lane storage evaluated in §9.0.

Intersection	Critical movement	Build Q95 (ft)
NW 107 Av & NW 50 St	Worst approach	934
Northwest 107th Avenue & Northwest 52nd Street	Worst approach	958
NW 52 St @ 10400 Blk	Worst approach	135
NW 50 St @ NW 10800 Blk	Worst approach	103
NW 52 St @ NW 10400 Blk	Worst approach	150
NW 109 Av @ NW 4800 Blk	Worst approach	97
NW 52 St @ NW 104 Av	Worst approach	126
NW 50 St @ NW 10900 Blk	Worst approach	88
NW 107 Av & NW 58 St	Worst approach	988
NW 50 St @ NW 11000 Blk	Worst approach	87
NW 58 St @ NW 10400 Blk	Worst approach	615

9.0 TURN LANE EVALUATION (2027)

Turn-lane evaluation proceeds in two steps per FDM Chapter 212: first whether a turn lane is warranted at each affected approach (a function of the turning volume, opposing/through volume, and speed), and second — where a bay is warranted or already exists — whether its storage length is adequate for the projected queue.

9.1 Turn-Lane Warrants

A turn-lane warrant is a movement-level screen: an exclusive left-turn lane is evaluated against the FDM Chapter 212 / NCHRP 745 left-turn-lane guidelines (a function of advancing volume, opposing volume, and posted speed), and an exclusive right-turn lane is conventionally warranted where the peak-hour right-turn volume exceeds roughly 40–60 vph (or the applicable local threshold, e.g., Miami-Dade County land-development code). The proposed development adds approximately 326 inbound and 354 outbound trips in the PM peak hour, distributed to the site driveways and adjacent intersections; the by-movement turning volumes required to apply the warrant thresholds are established from the AM/PM turning-movement counts and the approved trip-distribution at the methodology meeting. This screening tool does not decompose approach volumes into left/through/right movements, so a definitive turn-lane warrant determination — particularly at the site driveways connecting to the SHS — is deferred to the sealed submittal against the counted and assigned movement volumes.

9.2 Storage-Bay Adequacy

Where a turn bay is warranted or already present, its storage length must contain the Build-scenario 95th-percentile queue without spill-back into the adjacent through lane; a deficit indicates the bay is shorter than the projected queue and requires either a bay extension (where adjacent infrastructure permits) or a documented engineering finding that the spill-back is acceptable. Deceleration and taper lengths follow FDM Chapter 212.

Storage-bay adequacy: no intersection carries a field-measured existing storage length (`existingStorageFt`). Supply that field on each affected intersection record to enable the Required-vs-Existing-vs-Deficit turn-lane table per FDM Chapter 212 storage / taper standards. Turn-lane warrants at the site driveways must be evaluated against the assigned turning volumes at submittal.

10.0 CONCURRENCY ANALYSIS

Transportation concurrency was made optional statewide by HB 7207 (2011). For new development, the Development of Regional Impact (DRI) review track was further curtailed by SB 1216 / Ch. 2015-30 (which added F.S. 380.06(30) routing otherwise-DRI projects through the State Coordinated Review Process at F.S. 163.3184(4) in lieu of DRI), with cleanup completed by CS/CS/HB 1151 / Ch. 2018-158 — DRI is now a legacy branch retained only for amendments/abandonments of existing DRIs. Miami-Dade County review framework: Concurrency retained + Chapter 33E multimodal mobility impact fee. Controlling document(s): CDMP + Administrative Order 4-85 + Code Ch. 33-G (concurrency) + Ch. 33E (mobility fee). LOS standard: Per the Miami-Dade Comprehensive Development Master Plan (CDMP) Transportation Element; SHS LOS D urbanized per FDOT Policy 000-525-006 applies on State Highway System frontage. Per Florida Statutes §163.3180(5)(h)1.a., local governments must consult with FDOT whenever a Strategic Intermodal System (SIS) facility is expected to be impacted by a comprehensive-plan amendment.

Note. Miami-Dade conventions observed in published CDMP-amendment exemplars: Engineer's Certification page as the first section (before Executive Summary); Methodology Letter placed at Appendix C (not the canonical Appendix A); the report closes with three parallel-track end-chapters numbered 1.0–3.0 each (Concurrency Analysis / CDMP Analysis / Zoning Analysis). The renderer surfaces those conventions as required structure for a Miami-Dade submittal but does not auto-generate the three-track parallel content.

10.1 Roadway Segment Capacity — Generalized Service Volumes

Roadway-segment level of service is screened against the FDOT Quality/Level of Service Handbook v6.0 (August 2025) Generalized Service Volume Tables (GSVTs). Per Q/LOS v6.0 the peak-hour two-way service volume is keyed to FDM Chapter 200 context class (C1–C6, C2T) and through-lane count, with D = 0.55 statewide. The applicable context class and lane count for each segment are FDOT Roadway Characteristics Inventory (RCI) attributes (Feature 126 context class; lane count from the RCI flat file) and must be confirmed during the methodology meeting. This screening tool selects the GSVT row only where those inputs are supplied and otherwise defers the segment v/c rather than assume a class.

Arterial (signalized) peak-hour two-way service volumes (vph) — Q/LOS v6.0 App. B:

Context / lanes	LOS C	LOS D	LOS E
C3C (Suburban Commercial), 2-lane undivided	1,380	1,950	—
C4 (Urban General), 2-lane	1,310	1,710	—
C3C, 4-lane divided	2,760	3,290	—

Context / lanes	LOS C	LOS D	LOS E
C4, 4-lane divided	2,070	2,980	—
C3R (Suburban Residential), 4-lane divided	3,090	3,360	—
C3C, 6-lane divided	4,290	4,870	—
C4, 6-lane divided	3,850	4,560	—
C2T (Rural Town), 2-lane undivided	1,310	1,710	—
C5 (Urban Center), 4-lane	2,350	3,450	3,870
C5, 6-lane	2,560	4,850	5,650
C6 (Urban Core), 4-lane	—	2,710	3,490
C6, 6-lane	—	4,960	5,350

C6 LOS C is undefined in v6.0 (C6 facilities are neither planned nor designed for auto LOS C); cells marked "—" past LOS D are F at signal capacity. Material adjustment multipliers (one-way × 1.2; multilane without exclusive LT × 0.75; 2-lane undivided without exclusive LT × 0.80; non-State signalized × 0.90; exclusive RT lane × 1.05) and per-context K factors (C1/C2/C2T 8.5–10.5%; C3C/C3R/C4 7.5–9.5%; C5/C6 7.0–9.0%) apply per Q/LOS v6.0 Ch. 6. Freeway (Limited Access) GSVT LOS-D capacities: Urbanized 4/6/8-lane = 7,400 / 11,050 / 14,710 vph (AADT-equivalent 82,200 / 122,800 / 163,400); Rural 4-lane divided = 5,950 vph (AADT-equivalent 56,700).

Per-segment GSVT v/c is not computed for this run: a segment context class (C1–C6 / C2T) and through-lane count were not supplied. Provide them as inputs — or rely on the FDOT RCI data adapter once wired — to populate the roadway-segment LOS table against the service volumes above. Segment AADT available to the screen is listed in §4.0.

10.2 Site Access / Ingress-Egress

Per MTSIH 2024 §3.2 Table 5, driveway TIA-scoping is keyed to gross trips per day (including pass-by): Category A 1–20 vpd (single-family); B 21–600 (small multifamily / very small commercial); C 601–1,500 (small-mid retail / small office); D 1,501–4,000 (mid retail / mid office); E 4,001–15,000 (large retail / mixed-use); F 15,001–30,000 (very large mixed-use / mall); G ≥30,001 (regional mall). Pre-application meeting + traffic study are required for Categories C–G (>600 vpd including pass-by). A connection-permit change-of-use is additionally triggered per F.S. 335.182(3)(b) when trip generation increases by >25% AND >100 vpd vs. the existing use. Connection to the FDOT State Highway System requires a connection permit per Rule 14-96 F.A.C. (last amended April 2, 2023). Driveway spacing, median-opening spacing, and signal spacing are governed by the access-management class (Classes 1–7) assigned to the impacted SHS segment per Rule 14-97 F.A.C. and FDOT Procedure 525-030-155; the class is stored in the RCI as Feature 146 / ACMANCLS (codes 00–07; 99 = unclassified, interim standards in Rule 14-97.004(1) apply until assignment). Driveway geometry (W, R, F, Y, G, Driveway Length, S, I; Categories A–D in FDM Chapter 214, Categories E–F–G punt to FDM Chapter 212), turn-lane warrants, deceleration-lane length, and intersection sight distance must be designed to FDOT Design Manual (FDM 2026) standards; off-SHS connections on city/county facilities follow the Florida Greenbook. The access-management class for the impacted SHS facility should be confirmed against the FDOT-published Access Management TDA layer.

Three-track end-chapter convention (Miami-Dade): the formal CDMP-amendment submittal should additionally provide three parallel-track end-chapters (Concurrency Analysis / CDMP Analysis / Zoning Analysis), each independently numbered 1.0–3.0. The Concurrency track is reviewed against Code Ch. 33-G + Admin. Order 4-85; the CDMP track is reviewed against the adopted Transportation Element; the Zoning track is reviewed against the host municipal zoning ordinance. This screening tool does not auto-generate the three-track parallel content; the inputs required for each track must be coordinated with the Miami-Dade TPO and Miami-Dade County DTPW prior to submittal.

11.0 TRANSIT AND MOBILITY

11.1 Transit Service

No transit stops detected within 0.25 mi of the site via the live Transit.land / OSM Overpass query.

Verify against the controlling transit agency's published GTFS feed (e.g., Broward County Transit, Miami-Dade Transit, MARTA, LYNX, JTA, HART) at the methodology meeting. If transit-mode reduction is applied to trip generation, the supporting service must be cited and an alternative-mode trip reduction memo retained in the methodology letter (Appendix A or C per jurisdiction).

11.2 Internal Circulation and Multimodal

Internal site circulation, parking access, and service-vehicle pathways depend on the final site plan and are not included in this screening-level analysis. Internal queuing at the principal driveway should be evaluated for adequate storage between the SHS edge of pavement and the first internal conflict point per FDM guidance. Pedestrian and bicycle connectivity to the surrounding network and to the transit stops above should be confirmed against FDM Chapter 222/223 (sidewalk / bicycle facilities).

12.0 CRASH ANALYSIS

Site-radius crash query against the FDOT SSO Crashes (All) public FeatureServer returned no records within 0.5 mi over the available data window (FDOT public extract is comprehensive through 2018-2019; post-2019 records require Signal4 Analytics agency login). A formal submittal must include the 3-year Crash Analysis Reporting (CAR) extract for the affected intersections — this screening tool's public-data path cannot supply that for current years.

13.0 PRELIMINARY SIGNAL WARRANT ANALYSIS

A preliminary traffic-signal warrant screening is performed against the MUTCD (2009 Edition, FDOT-adopted per the Florida Traffic Engineering Manual) warrants — Warrant 1 (Eight-Hour Vehicular Volume), Warrant 2 (Four-Hour Vehicular Volume), Warrant 3 (Peak Hour), Warrant 4 (Pedestrian), Warrant 5 (School Crossing), Warrant 6 (Coordinated Signal System), Warrant 7 (Crash Experience), and Warrant 8 (Roadway Network). A definitive warrant determination requires the full 8-hour approach-volume counts and the applicable engineering study; a signal is not justified on any single warrant alone without an engineering study demonstrating that a signal will improve the overall safety and/or operation of the intersection.

Based on the projected Build-condition operations, the following study locations should carry a full signal / control-modification warrant analysis (Warrants 1–3 with 8-hour counts, plus Warrant 7 where the §12.0 crash review indicates a correctable pattern) at submittal:

Study location	Build LOS	Δ delay (s)	Preliminary indication
NW 107 Av & NW 50 St	F	0.0	At/over LOS E — monitor
Northwest 107th Avenue & Northwest 52nd Street	F	0.0	At/over LOS E — monitor
NW 107 Av & NW 58 St	F	0.0	At/over LOS E — monitor
NW 58 St @ NW 10400 Blk	F	0.0	At/over LOS E — monitor

Preliminary indication is a screening flag from projected LOS/delay only, not a warrant determination. The site driveways connecting to the SHS must additionally be evaluated for signalization / turn-lane control against the assigned turning volumes and the access-management class per Rule 14-97 F.A.C.

14.0 CONCLUSIONS AND RECOMMENDATIONS

The proposed Miami-Dade County Sample — Shopping Center / Retail is projected to generate 6,800 daily trips (326 in / 354 out in the PM peak hour) at full build-out. Under Build (Scenario 3) conditions at opening year 2027, 0 study intersections project to drop a Level of Service grade and 4 operate at LOS E or F. The improvement alternatives identified in §7.1 should be advanced, and the applicable proportionate-share / mobility-fee obligation should be calculated per the controlling Miami-Dade County ordinance to maintain the FDOT SHS LOS standard within the study network.

14.1 Findings

• Project will generate 6,800 new daily vehicle trips, with 680 during the PM peak hour (326 inbound / 354 outbound).

• Pass-by credit 30% and internal-capture credit 0% applied at the PM peak (25% of that credit at the AM and Saturday-midday periods, per the industry rule of thumb that off-peak shopping diverts less) before off-site assignment (standard pass-by methodology; ULI Internal Capture).

• Auto-mode share of 86% applied to external trips for Miami-Dade County; 14% of generated trips are assumed to arrive by transit, walking, or cycling and do not load the off-site roadway network. Source: ACS 5-Year Estimates Table B08301 (US) or national-stats equivalent.

• Existing volumes were grown by 1.50%/yr over 1 year (×1.01) to the opening-year horizon.

• 11 signalized intersections fall within the study area; 0 are projected to drop at least one LOS grade after build-out.

• 4 intersections are projected to operate at LOS E or F under the build condition and require formal mitigation per Miami-Dade Department of Transportation and Public Works TIS guidance.

• Conclusions are reported at the applied screening assumptions. Discrete-scenario sensitivity at the formal TIA scoping meeting should test: trip-generation method (rate vs. fitted-curve equation); internal capture and pass-by credit variants; and a ±0.5%/yr background growth band around the applied value.

14.2 Programmed Projects

Committed-projects review should consult the FDOT Five-Year Work Program (https://www.fdot.gov/workprogram) and the controlling MPO/TPO Transportation Improvement Program (TIP) and Long Range Transportation Plan (LRTP). Programmed roadway and intersection improvements within the study area should be incorporated into the No-Build network. This screening analysis does not automatically integrate Work Program data; manual review is recommended for any submittal.

14.3 Professional Engineer Certification

A Florida TIS / MTIA deliverable must be signed and sealed by a Florida-licensed Professional Engineer per Florida Statutes Chapter 471 and Florida Administrative Code Rule 61G15-23.001. The cover and signature page of the formal submittal must bear the seal, signature, and date of the Engineer of Record.

14.4 Methodology Notes

• Trip generation uses public-data average rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.

• Pass-by and internal-capture credits are applied in full at the PM peak (and at 25% of that credit fraction for the AM and Saturday-midday periods) per standard pass-by screening methodology and ULI Mixed-Use Internal Capture defaults; only the residual external vehicle trips are assigned to off-site intersections.

• Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.

• Weather adjustment follows HCM 6th-Edition Ch. 11 (rain/snow capacity reduction): clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.

• Off-site impact is screened for all signalized intersections within the study radius (default 0.5 mi) using the four-step travel demand model (FHWA; NCHRP Report 716). Step 1 Trip Generation: public-data average rates (SANDAG 2002 / NHTS 2017 / NCHRP 716) give the site's external (post pass-by / internal-capture) productions. Step 2 Trip Distribution: the Caltran mass/distance gravity model — the Florida distribution standard (Caltran Engineering HCA Westside TIS) — allocates trips to surrounding zones by $T_j = (M_j / (d_j \cdot d_{site})) / \Sigma(M_x / (d_x \cdot d_{site}))$, where mass M_j is each signal's through-volume (destination-activity attraction proxy) and d_j is its straight-line distance from the site (site-zone distance normalizer $d_{site} = 1$). The normalized zone shares set the directional distribution and drive the project-trip assignment. Step 3 Mode Choice: a binary logit $P(auto)=1/(1+e^{-(ASC-\lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (a density proxy from surrounding through-volumes) so denser, more transit-served sites split further from auto; only the resulting vehicle trips load the network. Step 4 Route Assignment: a capacity-constrained assignment using the BPR volume-delay function $t = t0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives. Signals lacking AADT data fall back to a constant 5,000 vpd attraction.

• After assignment, all signalized intersections within the study radius are reported in the affected-intersections table. Project-added trips, v/c ratio change, control delay change, LOS change, and 95th-percentile queue are reported for each intersection so the reviewer can assess relative impact. Intersections beyond the study radius are excluded from analysis.

- Auto-mode share is applied per metro before assignment. Suburban-US metros default to 90% auto (ACS 5-Year B08301 median); transit-heavy metros use measured auto-mode share (e.g., NYC 32%, Tokyo 30%, London 38%, San Francisco 47%). Non-auto trips (transit, walking, cycling) do not load the off-site roadway. This is a screening-level adjustment; a real TIS submittal in a transit-heavy market should refine with project-specific TAZ data.
- Candidate signals are de-duplicated within a 45m clustering threshold to prevent OSM divided-arterial splits and way-record artifacts from double-counting a single physical intersection.
- Intersection-level control delay uses the HCM signalized-intersection model d = d1 + d2 (Webster uniform delay + Akçelik/HCM incremental-delay term) with a 90s cycle, g/C = 0.45, 1,800 vphpl saturation flow (× weather factor), 15-minute peak analysis period (T = 0.25 hr) and pretimed-signal incremental-delay factor k = 0.5. Reported control delay is capped at 120 s (LOS F) for screening reliability — the incremental-delay term is not calibrated far above capacity, so oversaturated approaches are reported as LOS F rather than an implausible delay figure; a calibrated design-level analysis (HCS / Synchro) supersedes this screen.
- Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (deterministic per-signal allocation perturbed ±15% from a 30/25/25/20 base) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (HCM Eq. 19-50, Q95 ≈ Q1 × 1.65 × 25 ft/veh) are reported.
- Level of Service is assigned from HCM 6th-Edition signalized-intersection control-delay thresholds (Exhibit 19-8): A ≤10s, B ≤20s, C ≤35s, D ≤55s, E ≤80s, F >80s.
- Sensitivity is reported in narrative form per standard TIA practice: trip-generation method (rate vs. fitted-curve equation), discrete internal capture and pass-by credit variants, and a ±0.5%/yr growth-rate band around the applied value. The engine retains an internal stochastic-sensitivity routine (Box-Muller-perturbed trip rate and existing volume) for demo-mode diagnostics; it is not surfaced in the deliverable because TIA sensitivity is conducted through discrete scoping-agreed scenario variants, not statistical perturbation of unmeasured distributions.
- Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.
- Turbo-lane (continuous-green-T) screening flags signalized 3-leg T-intersections where one or more main-street through lanes could flow continuously while the minor-street left turn merges in the median — recovering the green time the through movement would otherwise lose to the signal. Candidacy requires measured 3-leg geometry, an arterial-class main street, and a detected median (divided carriageway), all derived from the OpenStreetMap road network. Approach-capacity gain is computed as (turbo lanes / approach lanes) × (1 − g/C) / (g/C), with the main-street through g/C derived from the modeled main-vs-minor critical-flow split; the result is reported within the +7%…+173% envelope documented in the continuous-green-T design literature — the standard national references for this geometry (David Plummer & Associates, Adding Turbo Lanes to T-Intersections, 2010; and the Design Guidelines for the Development of Continuous Green Intersections, 1997). Lane counts and signal timing must be field-verified before design.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

The Caltran mass/distance gravity model (the Florida distribution standard, §6.1) allocates trips to surrounding zones: T_j = (M_j / (d_j · d_site)) / Σ (M_x / (d_x · d_site)). Zone mass M_j is each signal's through-volume (destination-activity proxy) and d_j its distance from the site (d_site = 1). The shares below are the §6.1 distribution.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
NW 107 Av & NW 50 St	0.06	0.1	382	31.8%
Northwest 107th Avenue & Northwest 52nd Street	0.16	0.4	280	23.3%
NW 50 St @ NW 10800 Blk	0.28	0.7	127	10.6%
NW 109 Av @ NW 4800 Blk	0.31	0.7	108	9.0%
NW 50 St @ NW 10900 Blk	0.41	1.0	65	5.4%
NW 107 Av & NW 58 St	0.46	1.1	61	5.1%
NW 52 St @ 10400 Blk	0.22	0.5	54	4.5%
NW 50 St @ NW 11000 Blk	0.46	1.1	51	4.2%
NW 52 St @ NW 10400 Blk	0.30	0.7	36	3.0%
NW 58 St @ NW 10400 Blk	0.49	1.2	24	2.0%
NW 52 St @ NW 104 Av	0.34	0.8	15	1.2%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 86% applied via a binary logit P(auto) = 1 / (1 + e^-(ASC − λ·ΔGC)) calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 14% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function t = t0 · [1 + 0.15·(v/c)^4] iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 1.73. Per-signal v/c, delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

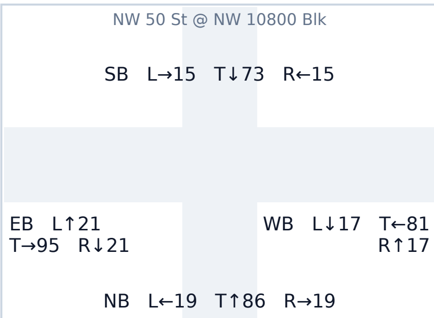
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 NW 50 St @ NW 10800 Blk

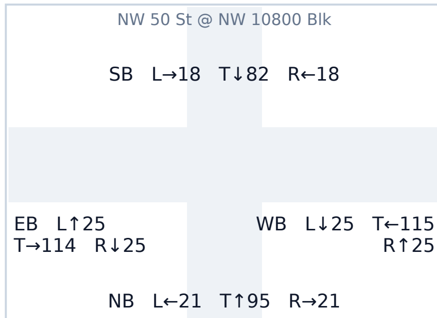
Signal ID	miami-dade-cdot-6072
Location	NE Miami-Dade · 0.28 mi from site
Added PM peak trips	127
Existing / No-Build (PM)	LOS B · 16.1 s/veh · v/c 0.25
Build (PM)	LOS B · 17.0 s/veh · v/c 0.32
Δ control delay	+0.9 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

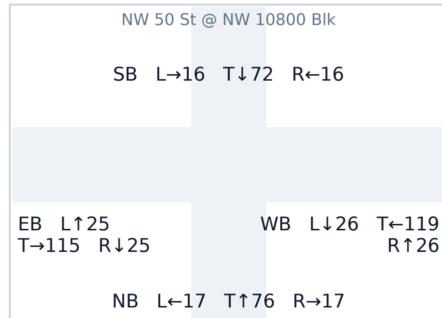
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	137	0	137	0.17 → 0.17	15.2 → 15.2	B → B	84
SB	104	14	118	0.13 → 0.15	14.8 → 14.9	B → B	71
EB	117	47	164	0.14 → 0.20	14.9 → 15.5	B → B	102
WB	99	66	165	0.12 → 0.20	14.7 → 15.6	B → B	103

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	51	40%
EB Through	47	37%
WB Right	15	12%
SB Left	14	11%

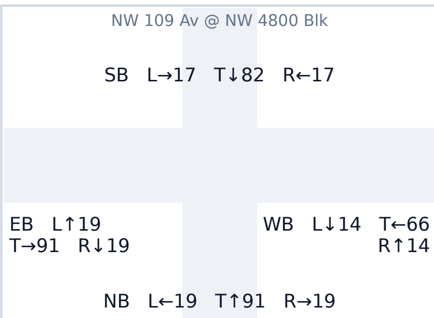
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 NW 109 Av @ NW 4800 Blk

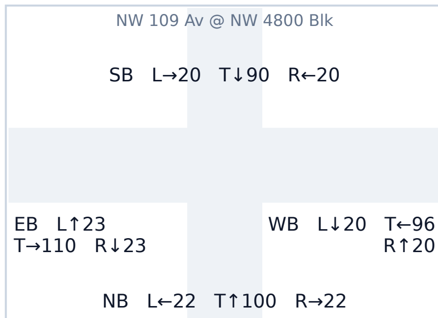
Signal ID	miami-dade-cdot-7519
Location	NE Miami-Dade · 0.31 mi from site
Added PM peak trips	108
Existing / No-Build (PM)	LOS B · 16.1 s/veh · v/c 0.25
Build (PM)	LOS B · 16.9 s/veh · v/c 0.31
Δ control delay	+0.8 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

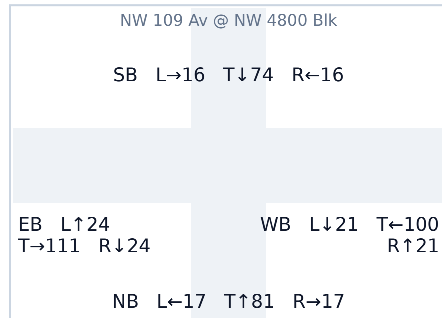
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	144	0	144	0.18 → 0.18	15.3 → 15.3	B → B	89
SB	126	4	130	0.16 → 0.16	15.0 → 15.1	B → B	79
EB	107	48	156	0.13 → 0.19	14.8 → 15.4	B → B	97
WB	80	56	136	0.10 → 0.17	14.5 → 15.2	B → B	83

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	52	48%
EB Through	48	44%
WB Right	4	4%
SB Left	4	4%

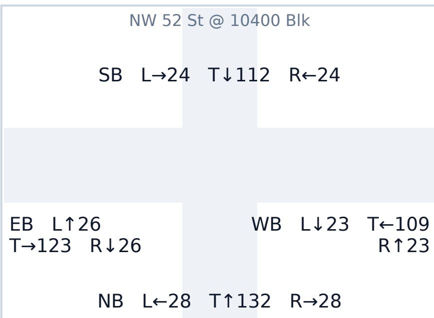
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 NW 52 St @ 10400 Blk

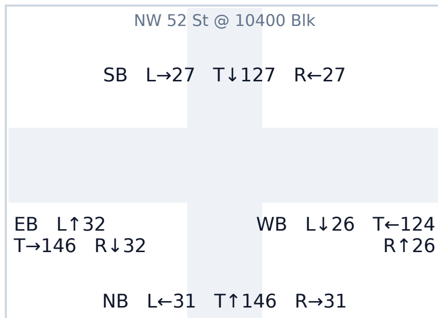
Signal ID	miami-dade-cdot-7490
Location	NE Miami-Dade · 0.22 mi from site
Added PM peak trips	54
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 18.6 s/veh · v/c 0.43
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

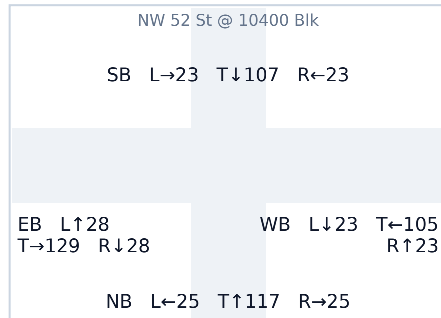
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	208	0	208	0.26 → 0.26	16.2 → 16.2	B → B	134
SB	169	12	181	0.21 → 0.22	15.6 → 15.8	B → B	114
EB	182	28	210	0.23 → 0.26	15.8 → 16.2	B → B	135
WB	162	14	176	0.20 → 0.22	15.5 → 15.7	B → B	111

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	15	28%
WB Through	14	26%
EB Left	13	24%
SB Right	12	22%

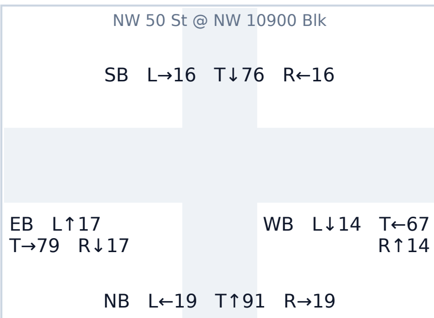
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 NW 50 St @ NW 10900 Blk

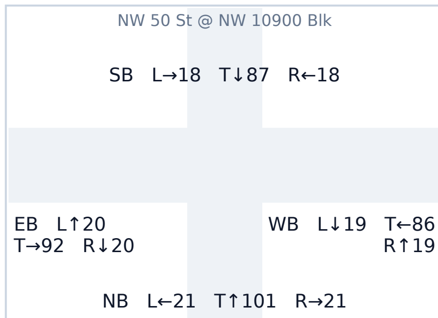
Signal ID	miami-dade-cdot-7270
Location	NE Miami-Dade · 0.41 mi from site
Added PM peak trips	65
Existing / No-Build (PM)	LOS B · 16.1 s/veh · v/c 0.25
Build (PM)	LOS B · 16.6 s/veh · v/c 0.29
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

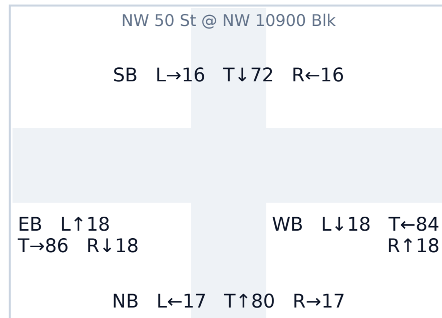
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	143	0	143	0.18 → 0.18	15.3 → 15.3	B → B	88
SB	114	9	123	0.14 → 0.15	14.9 → 15.0	B → B	75
EB	109	23	132	0.13 → 0.16	14.8 → 15.1	B → B	81
WB	91	33	124	0.11 → 0.15	14.6 → 15.0	B → B	76

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	24	37%
EB Through	23	35%
WB Right	9	14%
SB Left	9	14%

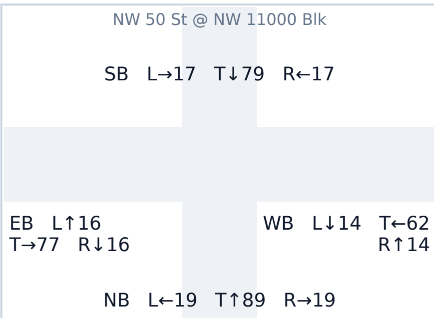
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 NW 50 St @ NW 11000 Blk

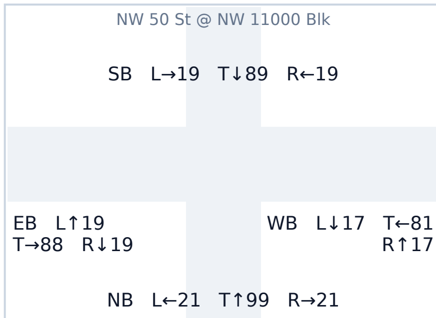
Signal ID	miami-dade-cdot-7269
Location	NE Miami-Dade · 0.46 mi from site
Added PM peak trips	51
Existing / No-Build (PM)	LOS B · 16.1 s/veh · v/c 0.25
Build (PM)	LOS B · 16.5 s/veh · v/c 0.28
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

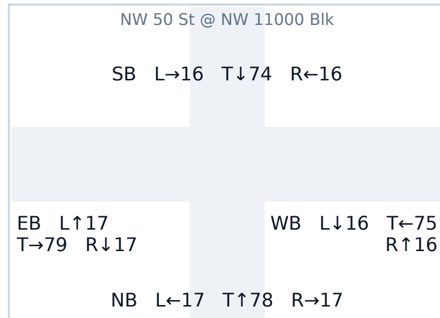
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	141	0	141	0.17 → 0.17	15.2 → 15.2	B → B	87
SB	120	6	127	0.15 → 0.16	15.0 → 15.1	B → B	77
EB	108	18	126	0.13 → 0.15	14.8 → 15.0	B → B	77
WB	88	27	115	0.11 → 0.14	14.6 → 14.9	B → B	69

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	20	39%
EB Through	18	35%
WB Right	7	14%
SB Left	6	12%

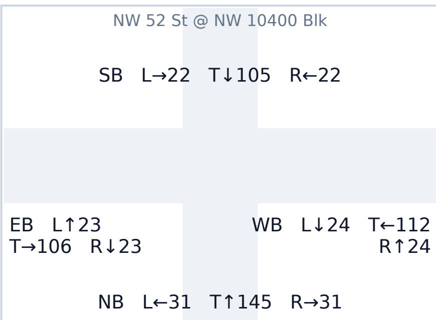
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 NW 52 St @ NW 10400 Blk

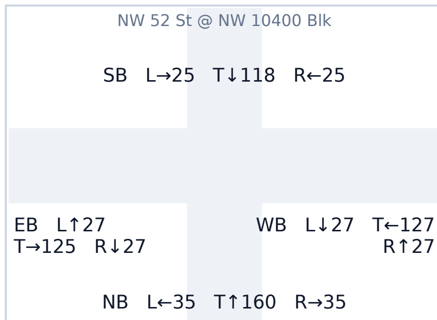
Signal ID	miami-dade-cdot-5777
Location	NE Miami-Dade · 0.30 mi from site
Added PM peak trips	36
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 18.4 s/veh · v/c 0.42
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

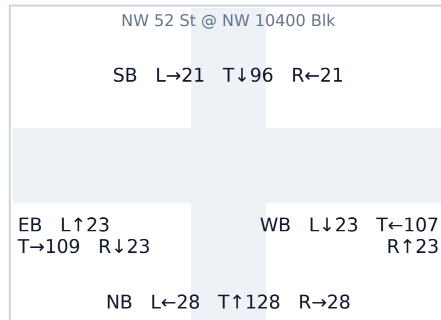
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	230	0	230	0.28 → 0.28	16.5 → 16.5	B → B	150
SB	162	6	168	0.20 → 0.21	15.5 → 15.6	B → B	105
EB	160	18	179	0.20 → 0.22	15.5 → 15.7	B → B	113
WB	170	12	181	0.21 → 0.22	15.6 → 15.8	B → B	114

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	12	33%
WB Through	12	33%
EB Left	6	17%
SB Right	6	17%

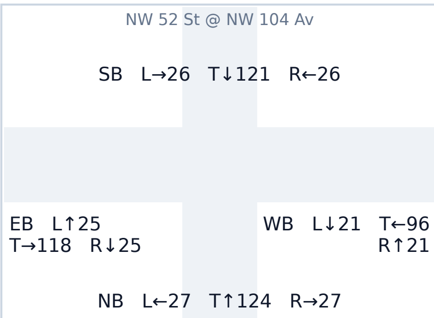
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 NW 52 St @ NW 104 Av

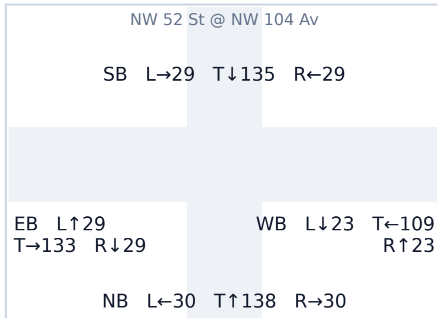
Signal ID	miami-dade-cdot-7489
Location	NE Miami-Dade · 0.34 mi from site
Added PM peak trips	15
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 18.2 s/veh · v/c 0.41
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

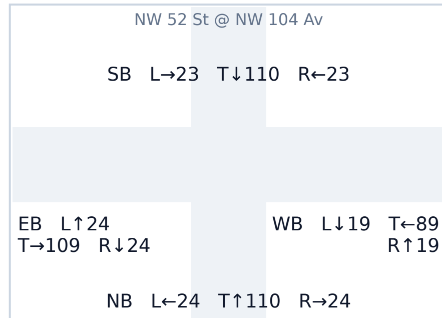
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	198	0	198	0.24 → 0.24	16.0 → 16.0	B → B	126
SB	191	2	193	0.24 → 0.24	15.9 → 15.9	B → B	123
EB	183	8	191	0.23 → 0.24	15.8 → 15.9	B → B	121
WB	150	5	155	0.19 → 0.19	15.4 → 15.4	B → B	96

Affected movements (PM peak project trips)

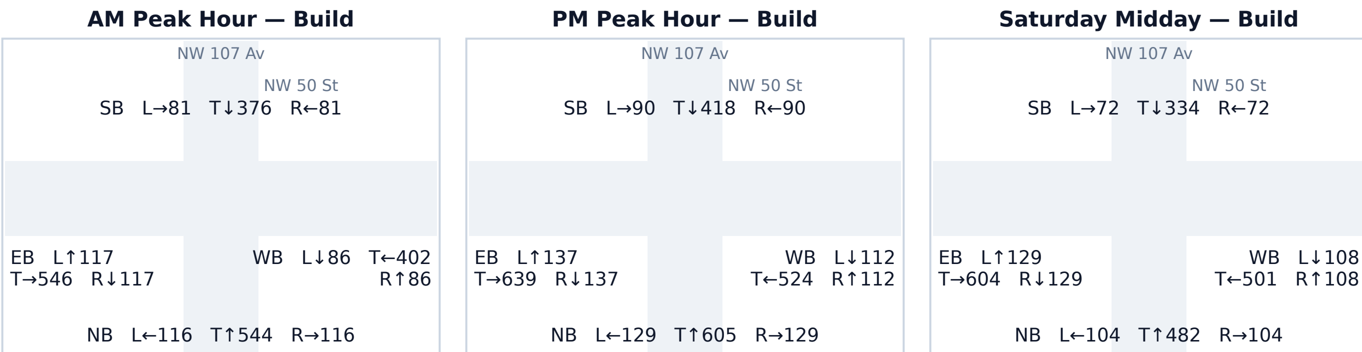
Movement	Project trips	% of project trips
EB Through	5	33%
WB Through	5	33%
EB Left	3	20%
SB Right	2	13%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 NW 107 Av & NW 50 St

Signal ID	miami-dade-cdot-8338
Location	NE Miami-Dade · 0.06 mi from site
Added PM peak trips	382
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.52
Build (PM)	LOS F · 120.0 s/veh · v/c 1.73
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	863	0	863	1.06 → 1.06	74.9 → 74.9	E → E	882
SB	598	0	598	0.74 → 0.74	26.3 → 26.3	C → C	508
EB	730	183	913	0.90 → 1.13	38.1 → 97.5	D → F	934
WB	550	199	748	0.68 → 0.92	24.2 → 41.1	C → D	727

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	199	52%
EB Through	183	48%

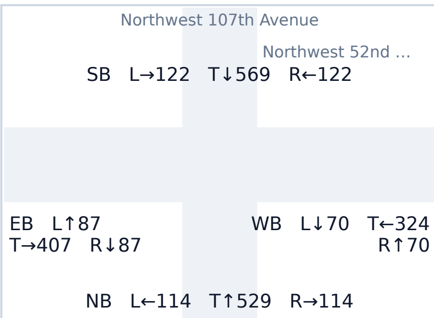
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Northwest 107th Avenue & Northwest 52nd Street

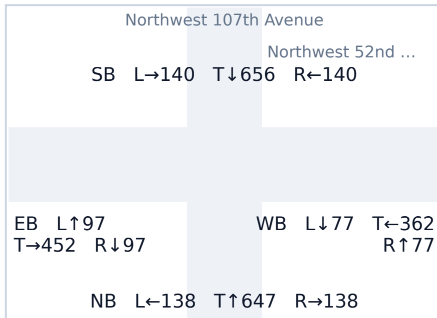
Signal ID	miami-dade-9175370051
Location	NE Miami-Dade · 0.16 mi from site
Added PM peak trips	280
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.52
Build (PM)	LOS F · 120.0 s/veh · v/c 1.68
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

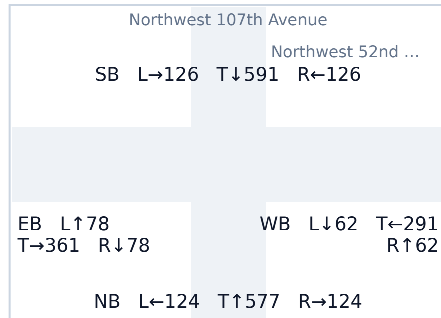
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	777	146	923	0.96 → 1.14	47.1 → 101.8	D → F	944
SB	804	132	936	0.99 → 1.16	54.5 → 108.4	D → F	958
EB	646	0	646	0.80 → 0.80	29.2 → 29.2	C → C	571
WB	514	2	516	0.63 → 0.64	22.8 → 22.9	C → C	410

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	143	51%
SB Through	132	47%
NB Right	3	1%
WB Left	2	1%

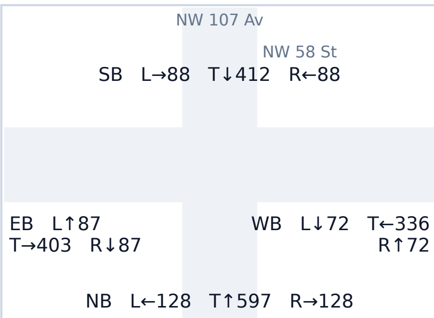
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 NW 107 Av & NW 58 St

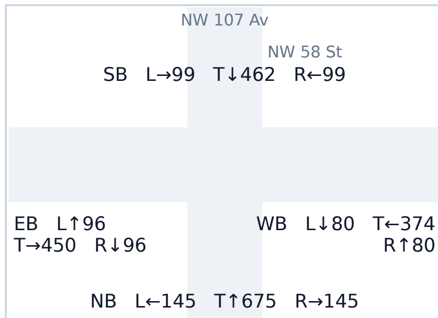
Signal ID	miami-dade-925462710
Location	NE Miami-Dade · 0.46 mi from site
Added PM peak trips	61
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.52
Build (PM)	LOS F · 120.0 s/veh · v/c 1.56
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

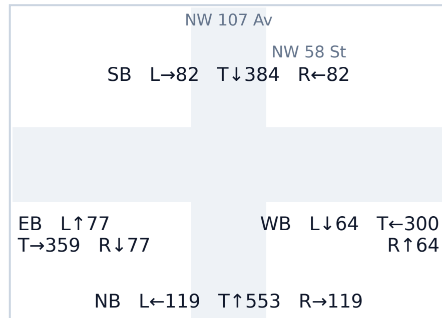
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	934	32	965	1.15 → 1.19	107.3 → 120.0	F → F	988
SB	633	28	660	0.78 → 0.82	28.4 → 30.4	C → C	592
EB	642	0	642	0.79 → 0.79	29.0 → 29.0	C → C	565
WB	533	1	534	0.66 → 0.66	23.5 → 23.5	C → C	431

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	30	49%
SB Through	28	46%
NB Right	2	3%
WB Left	1	2%

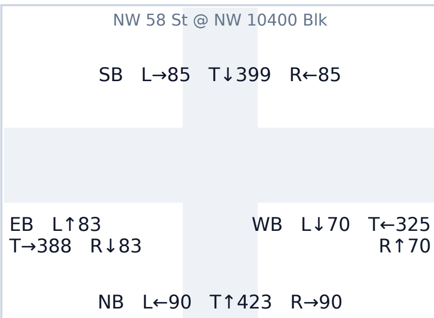
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 NW 58 St @ NW 10400 Blk

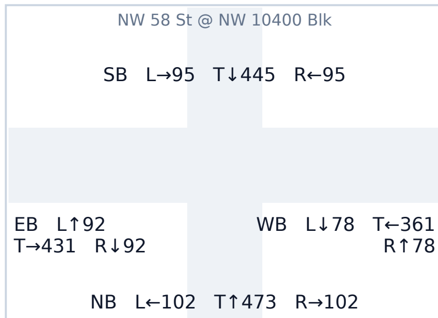
Signal ID	miami-dade-cdot-7450
Location	NE Miami-Dade · 0.49 mi from site
Added PM peak trips	24
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.34
Build (PM)	LOS F · 120.0 s/veh · v/c 1.36
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

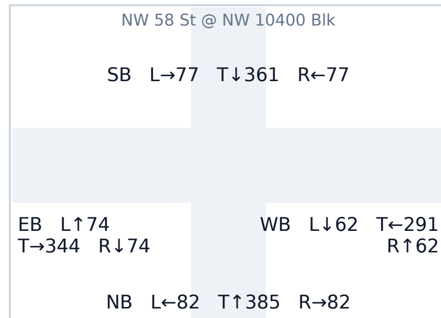
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	665	13	677	0.82 → 0.84	30.7 → 31.8	C → C	615
SB	625	10	635	0.77 → 0.78	27.9 → 28.5	C → C	556
EB	615	0	615	0.76 → 0.76	27.3 → 27.3	C → C	530
WB	516	1	517	0.64 → 0.64	22.9 → 22.9	C → C	412

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	11	46%
SB Through	10	42%
NB Right	2	8%
WB Left	1	4%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.